

Ribozyme-mediated inhibition of 801-bp deletion-mutant epidermal growth factor receptor mRNA expression in glioblastoma multiforme.

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The epidermal growth factor receptor (HER1/EGFR) is known to be dysregulated in a large subgroup of glioblastoma multiforme cases. Dysregulation of HER1/EGFR is related to malignant transformation and tumor growth in various human cancers, including malignant glioma. One mechanism that may lead to dysregulated HER1/EGFR signaling is the intrinsic alteration of the receptor structure due to mutational changes. The most common mutant form of HER1/EGFR, named variant III (EGFRvIII), results from an 801 bp in-frame deletion in the DNA sequence encoding the extracellular ligand-binding domain. Independent of ligand-binding, EGFRvIII is constitutively activated and beyond external control. Since its cellular expression was shown to relate enhanced tumorigenicity, various therapeutic strategies were developed to target EGFRvIII, including monoclonal antibodies, vaccination therapies and small-molecule tyrosine kinase inhibitors. In this review, we focus on ribozyme-mediated inhibition of EGFRvIII messenger RNA expression as a gene therapeutic approach for EGFRvIII-expressing glioblastoma multiforme.

Burden of severe maternal peripartum mental disorders in low- and middle-income countries: a systematic review.

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Peripartum severe mental disorders (PSMDs) encompass schizophrenia, affective psychosis, and psychotic and non-psychotic forms of bipolar disorders. PSMDs are well documented in high-income countries. However, much less is known about the prevalence of PSMDs in low- and middle-income countries (LMICs). The aim was to review the available literature systematically and estimate the prevalence of PSMDs among women in LMICs. We searched the Ovid MEDLINE, Embase, PsycINFO, CINAHL and Maternity and Infant Care databases systematically from the date of inception to Dec 31, 2020, for English-language publications with data on the prevalence of PSMDs among women in World Bank-defined LMICs. Selection of studies, extraction of data and assessment of study quality were each undertaken independently by at least two of the investigators. A total of five studies (completed in three countries spanning two continents) met the inclusion criteria. Five studies reported cumulative incidence of postpartum psychosis (ranging from 1.1 to 16.7 per 1000 births). We found no studies on the prevalence of severe mental disorder during pregnancy in these settings. Marked heterogeneity in methodology precluded meta-analysis. These findings indicate that PSMDs occur at a similar prevalence in low- and middle-income to high-income countries. However overall, there is a paucity of high-quality evidence from these settings. There is a need for rigorous studies with standardized methods to increase knowledge of the nature, prevalence, and determinants of PSMDs among women in resource-constrained LMICs to inform policies, service development, program planning and health professional training.

Experiences of Mental Health Care Among Women Treated for Postpartum Psychosis in England: A Qualitative Study.

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Postpartum psychosis has been found to affect 0.89-2.6 per 1000 women. Onset is typically rapid and

severe. Early recognition and appropriate treatment are crucial for a good prognosis. Our aim in this study was to understand women's experiences of mental health care and services for psychosis in the postnatal period. Semi-structured interviews were conducted with 12 women who reported being treated for postpartum psychosis. Findings were analysed thematically. Women reported that healthcare professionals across maternity and mental health services often lacked awareness and knowledge of postpartum psychosis and did not always keep them or their partners/families informed, supported, and involved. Women wanted better collaboration between and within services, and more efficient, appropriate, and timely care. They valued inpatient services that could meet their needs, favouring Mother and Baby Units over general psychiatric wards. Early Intervention in Psychosis services and specialist perinatal community mental health teams were also well liked.

Postpartum psychosis during delivery hospitalizations and postpartum readmissions, 2016-2019.

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Up-to-date data on population-level risk factors for postpartum psychosis is limited, although increasing substance use disorders, psychiatric disorders, autoimmune disorders, and other medical comorbidities in the obstetrical population may be contributing to the increased baseline risk of postpartum psychosis. This study aimed to determine trends in and risk factors for postpartum psychosis during delivery hospitalizations and postpartum readmissions. Analyzing the 2016-2019 Nationwide Readmission Database, this repeated cross-sectional study identified diagnoses of postpartum psychosis during delivery hospitalizations and postpartum readmissions within 60 days of discharge. The relationship among demographic, clinical, and hospital-level factors present at delivery and postpartum psychosis was analyzed with logistic regression models with adjusted odds ratios with 95% confidence intervals as measures of association. Separate models were created for postpartum psychosis diagnoses at delivery and during postpartum readmission. Temporal trends in diagnoses were analyzed with Joinpoint regression to determine the average annual percent change with 95% confidence intervals. Of 12,334,506 deliveries in the analysis, 13,894 (1.1 per 1000) had a diagnosis of postpartum psychosis during the delivery hospitalization, and 7128 (0.6 per 1000) had a 60-day postpartum readmission with postpartum psychosis. Readmissions with postpartum psychosis increased significantly during the study period ($P=.046$). Most readmissions with a postpartum psychosis diagnosis occurred in 0 to 10 days (43% of readmissions) or 11 to 20 days (18% of readmissions) after discharge. Clinical factors with the highest adjusted odds for postpartum psychosis readmission included delivery postpartum psychosis (adjusted odds ratio, 5.8; 95% confidence interval, 4.2-8.0), depression disorder (adjusted odds ratio, 3.7; 95% confidence interval, 3.3-4.2), bipolar spectrum disorder (odds ratio, 2.9; 95% confidence interval, 2.3-3.5), and schizophrenia spectrum disorder (adjusted odds ratio, 2.9; 95% confidence interval, 2.1-4.0). In models analyzing postpartum psychosis diagnoses at delivery, risk factors associated with the highest odds included anxiety disorder (adjusted odds ratio, 3.9; 95% confidence interval, 3.5-4.2), schizophrenia spectrum disorder (adjusted odds ratio, 2.5; 95% confidence interval, 1.9-3.4), bipolar disorder (adjusted odds ratio, 1.8; 95% confidence interval, 1.6-2.1), stillbirth (odds ratio, 3.6; 95% confidence interval, 3.1-4.2), and substance use disorder (odds ratio, 1.7; 95% confidence interval, 1.6-1.9). In addition, chronic conditions, such as pregestational diabetes mellitus, obesity, and substance use, were associated with delivery and readmission postpartum psychosis. This study determined that postpartum psychosis is increasing during postpartum readmissions and is associated with a wide range of obstetrical and medical comorbidities. Close follow-up care after delivery for other medical and obstetrical diagnoses may represent an opportunity to identify postpartum psychiatric conditions,

including postpartum psychosis.

The global prevalence of postpartum psychosis: a systematic review.

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Mental health is a significant contributor to global burden of disease and the consequences of perinatal psychiatric morbidity can be substantial. We aimed to obtain global estimates of puerperal psychosis prevalence based on population-based samples and to understand how postpartum psychosis is assessed and captured among included studies. In June 2014, we searched PubMed, CiNAHL, EMBASE, PsycINFO, Sociological Collections, and Global Index Medicus for publications since the year 1990. Criteria for inclusion in the systematic review were: use of primary data relevant to pre-defined mental health conditions, specified dates of data collection, limited to data from 1990 onwards, sample size >200 and a clear description of methodology. Data were extracted from published peer reviewed articles. The search yielded 24,273 publications, of which six studies met the criteria. Five studies reported incidence of puerperal psychosis (ranging from 0.89 to 2.6 in 1000 women) and one reported prevalence of psychosis (5 in 1000). Due to the heterogeneity of methodologies used across studies in definitions and assessments used to identify cases, data was not pooled to calculate a global estimate of risk. This review confirms the relatively low rate of puerperal psychosis; yet given the potential for serious consequences, this morbidity is significant from a global public health perspective. Further attention to consistent detection of puerperal psychosis can help provide appropriate treatment to prevent harmful consequences for both mother and baby.

Familial risk of postpartum psychosis.

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Postpartum psychosis, a mood disorder triggered by childbirth, is one of the most severe psychiatric conditions, with high risks of suicide and infanticide if untreated. While it is evident that genetic factors play a crucial role in disorder risk, the exact extent of their importance is yet to be determined. This cohort study consisted of 1,648,759 women from the Swedish nationwide registers, of whom 2,514 (0.15%) experienced postpartum psychosis within three months of their first-ever childbirth. We estimated the relative recurrence risk of postpartum psychosis for female full siblings and cousins as a measure of familial, genetic, and environmental risk. Relative recurrence risk of postpartum psychosis in full siblings was 10.69 (95% CI=6.60-16.26) when adjusted for year of and age at childbirth. Although cousins showed an elevated relative recurrence risk, these results did not reach statistical significance (1.78, 95% CI=0.70-3.62). Despite the higher familial risk of postpartum psychosis among full siblings, the absolute risk for women with an affected sibling is relatively low, estimated at 1.55% within the entire population. The observed increased risk of postpartum psychosis in full siblings suggests both genetic and shared environmental influences. However, the lack of significant results in cousins hampers a definitive distinction between these factors. Furthermore, despite increased relative recurrence risk in siblings, their overall likelihood of developing postpartum psychosis remains low. Our study underscores the need for further research to better understand the intricate interplay of genetics and environment in the development of postpartum psychosis.

Phenomenology, Epidemiology and Aetiology of Postpartum Psychosis: A Review.

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Postpartum psychoses are a severe form of postnatal mood disorders, affecting 1-2 in every 1000 deliveries. These episodes typically present as acute mania or depression with psychosis within the first few weeks of childbirth, which, as life-threatening psychiatric emergencies, can have a significant adverse impact on the mother, baby and wider family. The nosological status of postpartum psychosis remains contentious; however, evidence indicates most episodes to be manifestations of bipolar disorder and a vulnerability to a puerperal trigger. While childbirth appears to be a potent trigger of severe mood disorders, the precise mechanisms by which postpartum psychosis occurs are poorly understood. This review examines the current evidence with respect to potential aetiology and childbirth-related triggers of postpartum psychosis. Findings to date have implicated neurobiological factors, such as hormones, immunological dysregulation, circadian rhythm disruption and genetics, to be important in the pathogenesis of this disorder. Prediction models, informed by prospective cohort studies of high-risk women, are required to identify those at greatest risk of postpartum psychosis.

Incidence of hospitalization for postpartum psychotic and bipolar episodes in women with and without prior prepregnancy or prenatal psychiatric hospitalizations.

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Postpartum psychosis occurs in 1 to 2 cases per 1000 live births. Most studies have not distinguished postpartum psychosis from bipolar disorder or the proportion of the incidence attributable to prepregnancy psychiatric morbidity. To determine the incidence of postpartum psychosis and bipolar disorder attributable to previous psychiatric hospitalization. Population-based study using linked registry data to determine postpartum onset of psychotic and bipolar episodes within 90 days after the first birth, by women with and without prepregnancy or prenatal psychiatric hospitalization. We assessed the type, number, and recency of previous hospitalizations on the incidence of hospitalization for postpartum psychotic and bipolar episodes. Nationwide Swedish Hospital Discharge and Medical Birth registers. Swedish women delivering a first live infant between January 1, 1987, and December 31, 2001. Postpartum hospitalization for psychosis or bipolar disorder. The cumulative incidences for postpartum psychotic and bipolar episodes (adjusted for age at first birth) were 0.07% and 0.03%, respectively. The incidence of psychiatric hospitalizations for postpartum psychotic or bipolar episodes among women without previous psychiatric hospitalizations was 0.04% and 0.01% of first births, respectively; for women with any psychiatric hospitalization before delivery, the incidence was 9.24% and 4.48%, respectively. For postpartum psychotic and bipolar episodes, the risk increased significantly with the recency of prepregnancy hospitalizations, number of previous hospitalizations, and length of most recent hospitalization. More than 40% of women hospitalized during the prenatal period for a bipolar or a psychotic condition were hospitalized again during the postpartum period. Approximately 90% of all postpartum psychotic and bipolar episodes occurred within the first 4 weeks after delivery. Almost 10% of women hospitalized for psychiatric morbidity before delivery develop postpartum psychosis after their first birth. This underscores the need for obstetricians to assess history of psychiatric symptoms and, with pediatric and psychiatric colleagues, to optimize the treatment of mothers with psychiatric diagnoses through childbirth.

The lived experiences of individuals with postpartum psychosis: A qualitative analysis.

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Postpartum psychosis (PP) is a severe psychiatric illness that occurs in about 1 to 2 per 1000 people in the perinatal period. To date, qualitative research investigating PP has focused on specific topics, such as treatment experiences or the impact of the illness on patients' lives and families. These studies have included small samples of women with histories of PP, often limited to certain geographical areas or treatment centers. Given the heterogeneity in presentations of PP and access to care, larger and geographically diverse samples are needed to broadly understand this complex illness. Initiated in 2018, the Massachusetts General Hospital Postpartum Psychosis Project (MGHP3) consists of a large, international sample of those who have experienced PP. In addition to the specific aims of MGHP3, which include to better understand the phenomenology and potential genetic underpinnings of PP, this investigation invites participants to qualitatively describe their narratives of postpartum psychosis. This analysis included 130 participants who reported on 133 episodes of PP. Participants' responses to the PP narrative prompt fell under several overarching categories: 1) broad psychosocial experiences surrounding postpartum psychosis, 2) impact on the mother-baby dyad, 3) treatment experiences, and 4) recovery experiences. Our findings shed light on a range of ways in which individuals' lives are impacted by this illness, and point to areas for future research and clinical directions to improve the support and care for individuals with PP and their families.

[Post-partum psychosis].

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Postpartum psychosis occurs in 1-2/1000 childbearing women. It is commonly admitted that it belongs to bipolar disorder with psychotic features. A strong link between puerperal psychosis and bipolar disorder has been established. Symptoms include rapid mood fluctuations, confusion, delusions, hallucinations and bizarre behaviour. It can lead to devastating consequences. It is a psychiatric emergency that requires an urgent evaluation to exclude any organic cause. Therefore, early identification and appropriate treatment are critical. A quick and effective relief is necessary for maternal and child health and mother-infant relationship. Perinatal health professionals have to be accurate screening postpartum psychosis symptoms and have to educate patients and their family.
